

Package #140

Implement wireless vibration and temperature sensors on draft gear mechanisms and journal bearings with onboard analytics

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'rolling-stock mechanical/electrical equipment failure' (FRA cause family E): 3 of 25 FRA Form 54 accidents in the retrieved window stem from equipment causes. Causes observed: 'Draft gear/mechanism broken or defective (including yoke)'; 'Journal (roller bearing) failure from overheating'; 'Other coupler and draft system defects, (CAR) (Provide detailed description in narrative)'; speeds 7–37 mph. LIMITS STAMPED INLINE: carrier-reported; cause codes are classifications, not proven root causes; above-threshold events only (Art. XXI.5 analog).

Mechanism: Condition monitoring using sensors and data processing

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.